SDU University

Exploratory Data Analysis and Regression Modeling of Traffic Accident Distance

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Contents

1 Introduction **2**

2 Exploratory Data Analysis **2**

2.1 Missing Values 2

2.2 Numerical Variables 3

2.3 Target Variable 5

2.4 Outlier Analysis 6

3 Regression Modeling **7**

3.1 Model 1 (M1): Linear Weather Model 7

3.2 Model 2 (M2): Polynomial Weather Effects 8

3.3 Model 3 (M3): Weather + State + Weather Condition 8

3.4 Model 4 (M4): Weather + Infrastructure 9

3.5 Model 5 (M5): Full Linear Model 10

3.6 Model 6 (M6): Log-Transformed Response 12

3.7 Model 7 (M7): Log Response with State and Weather Conditions 13

3.8 Model 8 (M8): Log Response with Polynomial Weather Effects 14

3.9 Model 9 (M9): Mixed Transformations Model 15

4 Model Comparison **16**

4.1 Confidence Intervals for the Final Model 16

5 Conclusion: Interpretation and Data Limitations **17**

1 Introduction

Traffic accidents constitute a major public safety concern, leading to injuries, fatalities, infrastructure damage, traffic congestion, and economic losses. While accident severity is frequently studied, the **distance of roadway affected by an accident** provides additional insight into traffic disruption, response complexity, and road network performance.

This report presents an exploratory data analysis (EDA) followed by a comprehensive regression modeling framework applied to traffic accidents recorded in the United States in 2023. The primary response variable is *Distance.mi.*, representing the length of roadway affected by an accident in miles.

The dataset consists of **146,923 observations** and **46 original variables** describing weather conditions, geographic location, and road infrastructure characteristics.

2 Exploratory Data Analysis

2.1 Missing Values

```
> colSums(is.na(data))
                   ID                Source              Severity            Start_Time
                    0                     0                     0                     0
             End_Time             Start_Lat             Start_Lng               End_Lat
                    0                     0                     0                     0
              End_Lng          Distance.mi.           Description                Street
                    0                     0                     0                     0
                 City                County                 State               Zipcode
                    0                     0                     0                     0
              Country              Timezone          Airport_Code     Weather_Timestamp
                    0                     0                     0                     0
       Temperature.F.         Wind_Chill.F.           Humidity...          Pressure.in.
                 3654                  4916                  4007                  3384
       Visibility.mi.        Wind_Direction       Wind_Speed.mph.     Precipitation.in.
                 4106                     0                  4636                  6717
    Weather_Condition               Amenity                  Bump              Crossing
                    0                     0                     0                     0
            Give_Way              Junction               No_Exit               Railway
                    0                     0                     0                     0
          Roundabout               Station                  Stop       Traffic_Calming
                    0                     0                     0                     0
       Traffic_Signal          Turning_Loop        Sunrise_Sunset        Civil_Twilight
                    0                     0                     0                     0
    Nautical_Twilight Astronomical_Twilight
                    0                     0
```

Figure 1: Missing values per variable.

Missing numerical weather variables were imputed using the **median**, which is robust to skewness and extreme observations.

2.2 Numerical Variables

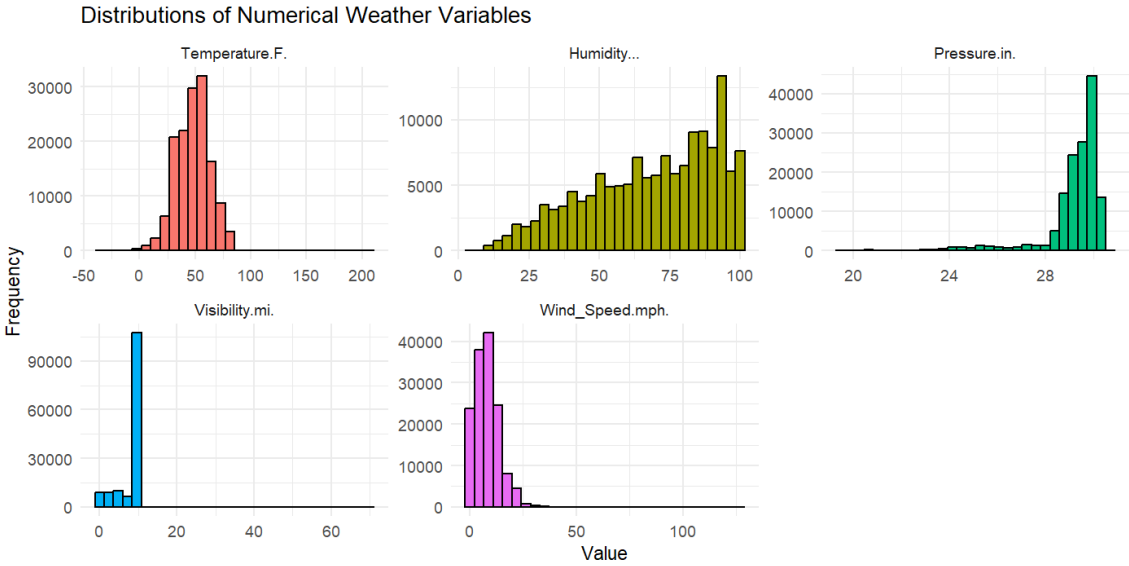

Figure 2: Distributions of numerical weather variables.

Figure 2 presents the empirical distributions of the numerical weather variables included in the analysis: temperature, humidity, atmospheric pressure, visibility, and wind speed. The histograms indicate substantial departures from normality, with pronounced skewness and uneven dispersion across variables. These distributional characteristics have important implications for data preprocessing and regression modeling.

Temperature (°F) exhibits an approximately unimodal distribution centered around moderate values, with the majority of observations falling between roughly 30 °F and 80 °F. While the distribution is fairly concentrated, it is not symmetric and shows a noticeable spread toward higher temperatures, reflecting occasional warm-weather accident conditions.

Humidity (%) displays a wide distribution over its bounded range from 0 to 100, with frequencies increasing toward higher humidity levels. This indicates that a large proportion of accidents occur under moderately to highly humid conditions. The bounded nature of the variable inherently limits normality and contributes to its asymmetric shape.

Atmospheric pressure (inches of mercury) is tightly clustered within a narrow interval near typical sea-level pressure values. The strong concentration and limited variability reflect physical constraints on atmospheric pressure, suggesting that although statistically detectable, its practical explanatory contribution may be limited.

Visibility (miles) shows a highly uneven distribution with a pronounced concentration at higher visibility levels and relatively few observations at low visibility. This pattern suggests that most accidents occur under clear visibility conditions, while reduced-visibility events are comparatively rare but still present.

Wind speed (mph) is strongly right-skewed, with most observations concentrated at low wind speeds and a rapidly declining frequency as wind speed increases. High-wind conditions appear infrequently and form a long right tail in the distribution.

Overall, the numerical weather variables exhibit non-normal distributions characterized by

skewness, boundedness, and strong concentration in specific ranges. These properties justify the use of median imputation for missing values and motivate the exploration of transformations and nonlinear specifications in subsequent regression models. The observed distributional patterns also suggest that simple linear relationships between individual weather variables and accident distance are likely to be weak, consistent with the modest explanatory power observed in later modeling results.

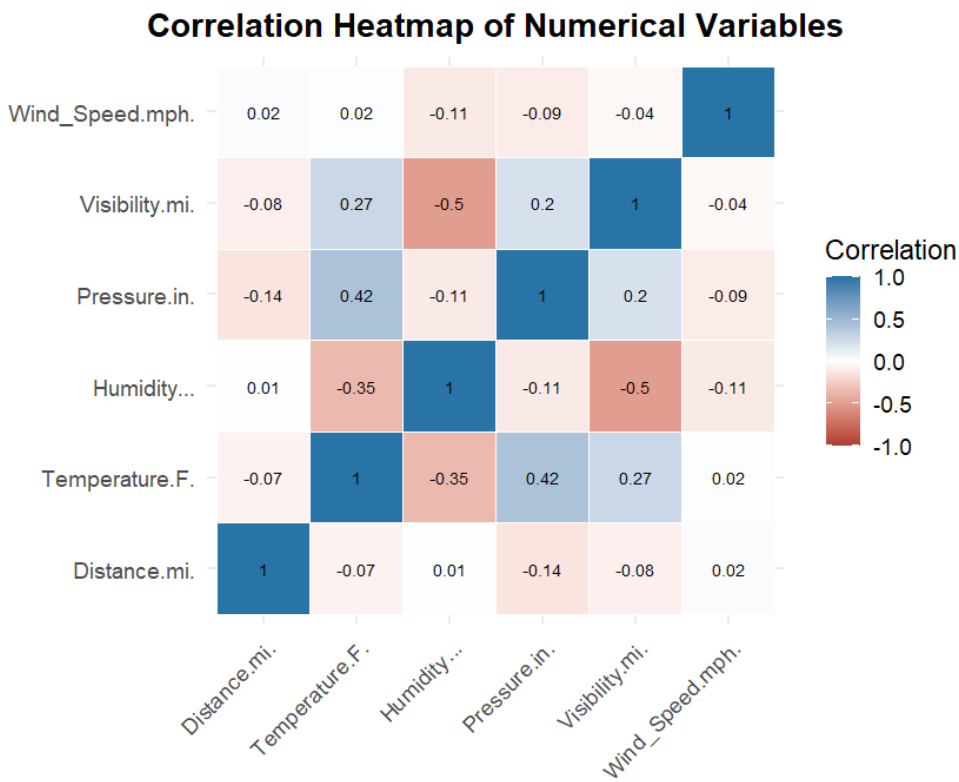

Figure 3: Correlation matrix of numerical variables.

Figure 3 presents the pairwise correlations among the numerical variables that will beused in the regression analysis. Overall, correlations are moderate to weak, indicating limited multicollinearity among predictors. Weather variables such as temperature, humidity, pressure, visibility, and wind speed exhibit expected relationships, but none show excessively strong linear dependence.

The response variable, *Distance*, displays weak correlations with all numerical predictors. This supports the regression results observed across models, where individual weather variables are statistically significant but explain only a small fraction of the total variability in accident distance.

2.3 Target Variable

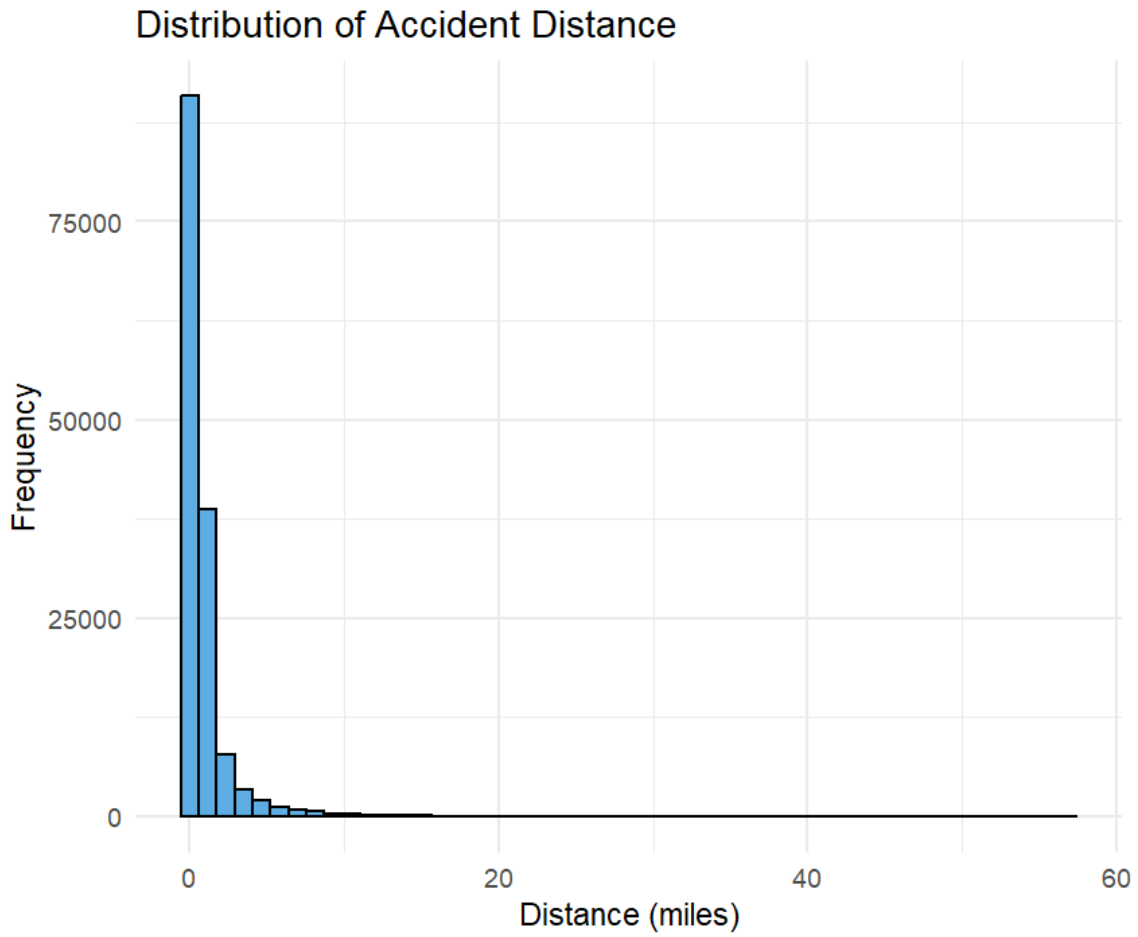

Figure 4: Distribution of accident distance (miles).

Figure 4 illustrates the distribution of accident distance, which is highly right-skewed with a large concentration of short-distance incidents and a small number of extreme values. Despite this skewness, accident distance is a continuous outcome that directly measures the physical extent of an accident's impact on the road network. This makes it a natural response variable for regression analysis compared to weather or infrastructure variables, which primarily describe conditions rather than outcomes. The observed distribution motivates the use of transformations and nonlinear specifications explored in later models, rather than excluding distance as an outcome of interest.

2.4 Outlier Analysis

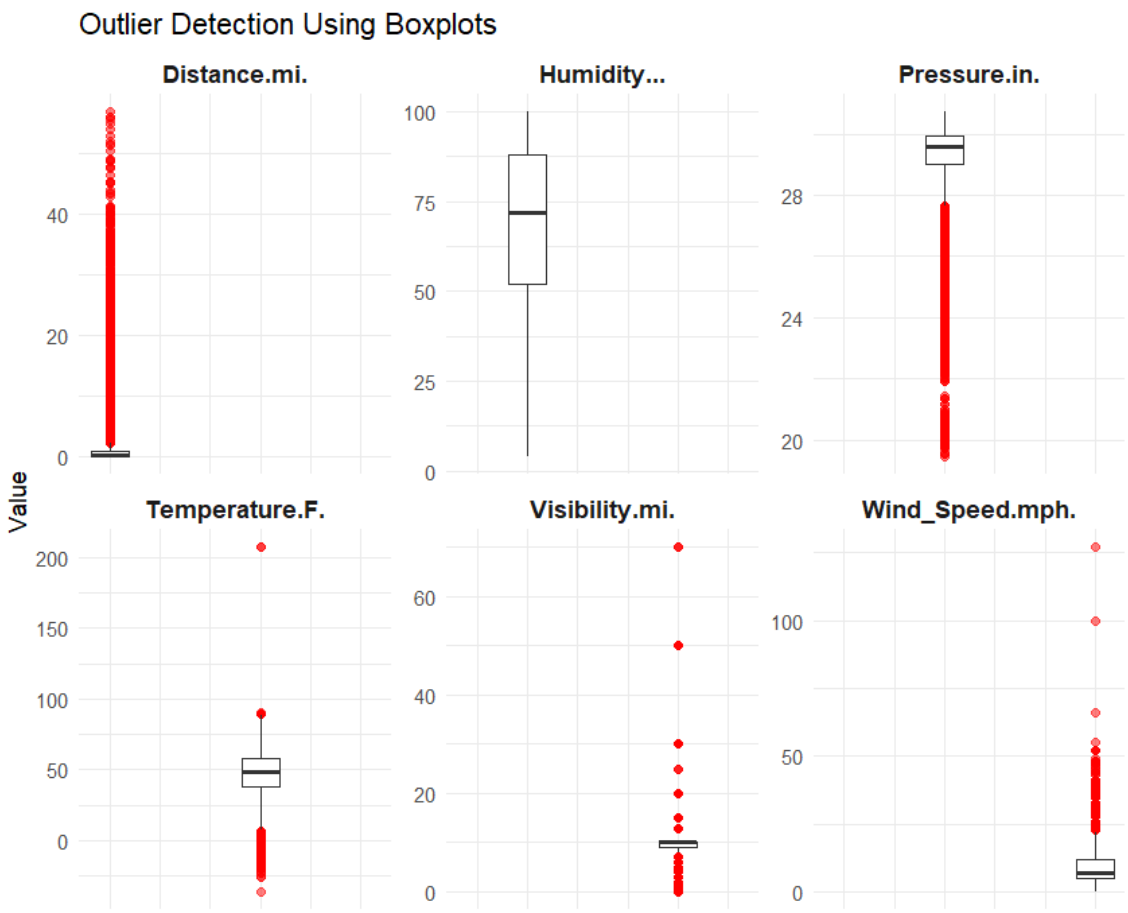

Figure 5: Outlier detection using boxplots.

Figure 5 illustrates the distributional properties of the response variable and key numerical predictors using boxplots. The position of each box relative to the axis reflects the skewness of the variable. Variables such as *Distance*, *Wind Speed*, and *Visibility* exhibit strong right-skewness, resulting in boxes concentrated near lower values with long upper whiskers and numerous points beyond the upper quartile. This pattern indicates that most observations are small, while a limited number of events take unusually large values.

In contrast, variables such as *Temperature* and *Pressure* display more compact boxes, reflecting lower variability under normal conditions. Occasional extreme values appear on either side of the distribution and correspond to rare but plausible weather situations, such as heat waves or abnormal pressure systems.

The relative size of each box represents the interquartile range (IQR), which captures the spread of the middle 50% of observations. Larger boxes indicate greater natural variability, while smaller boxes suggest more stable measurements. The presence of asymmetric whiskers further highlights departures from symmetry and reinforces the non-normal nature of several predictors.

Outliers identified in these boxplots were not removed from the dataset. These observa-

tions represent legitimate real-world events rather than measurement errors, particularly in the context of traffic accidents and extreme weather. Removing them would distort the underlying data-generating process and reduce the model's ability to capture rare but impactful scenarios. Instead, their influence was addressed through robust imputation, transformation, and alternative model specifications explored in later sections.

3 Regression Modeling

3.1 Model 1 (M1): Linear Weather Model

$$\text{Distance} = \beta_0 + \beta_1 T + \beta_2 H + \beta_3 P + \beta_4 V + \beta_5 W + \varepsilon$$

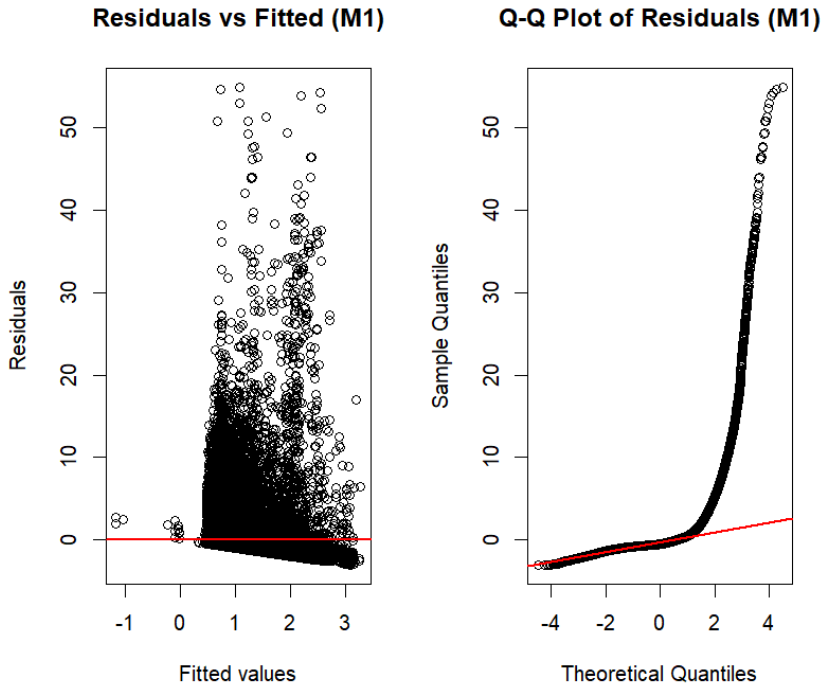

Figure 6: Residual diagnostics for Model 1.

Model 1 estimates the relationship between accident distance and numerical weather variables using ordinary least squares. All predictors are statistically significant at the 5% level based on individual t-tests. Temperature, humidity, atmospheric pressure, and visibility have negative coefficients, indicating that higher values of these variables are associated with slightly shorter accident distances, holding other factors constant. Wind speed exhibits a positive coefficient, suggesting a weak positive association with accident distance.

The t-tests assess the null hypothesis $H_0 : \beta_j = 0$ for each coefficient. Given the very large sample size ($n = 146{,}923$), even small effect sizes yield large t-statistics and high statistical significance, which explains the widespread rejection of the null hypotheses.

The overall F-test is highly significant ($F = 692.3$, $p < 0.001$), rejecting the joint null hypothesis that all slope coefficients are equal to zero. This confirms that the weather variables

collectively improve model fit relative to an intercept-only model. However, the low R^2 value ($R^2 = 0.023$) indicates that the model explains only a small proportion of the variability in accident distance, highlighting limited practical explanatory power.

3.2 Model 2 (M2): Polynomial Weather Effects

$$\text{Distance} = \beta_0 + \sum \beta_i X_i + \sum \gamma_i X_i^2 + \varepsilon$$

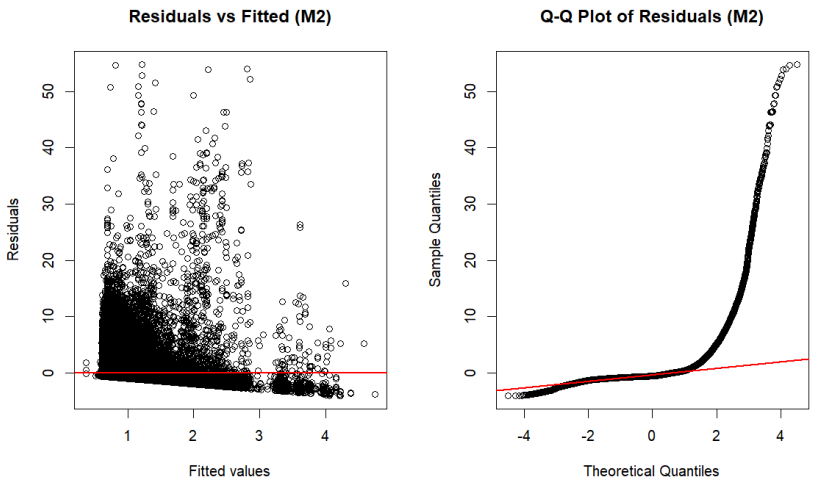

Figure 7: Residual diagnostics for Model 2.

Model 2 extends the baseline specification by including quadratic terms for all numerical weather variables in order to capture potential nonlinear relationships with accident distance. The linear components of temperature, humidity, pressure, and visibility remain statistically significant and negative, while wind speed is not significant in its linear form.

Several quadratic terms are statistically significant based on individual t-tests, including temperature, humidity, pressure, visibility, and wind speed squared. The positive coefficients on the squared terms indicate weak convex relationships, suggesting diminishing or reversing marginal effects at higher values of the predictors. However, the magnitudes of these nonlinear effects are small.

The overall F-test is highly significant ($F = 374.7$, $p < 0.001$), rejecting the joint null hypothesis that all slope coefficients are equal to zero. Despite this, the explanatory power improves only marginally relative to Model 1, with R^2 increasing from approximately 0.023 to 0.025. This indicates that nonlinear weather effects exist but contribute little additional explanatory value in practical terms.

3.3 Model 3 (M3): Weather + State + Weather Condition

$$\text{Distance} = \beta_0 + \sum_{j=1}^{p} \beta_j X_j + \sum_{k=1}^{q} \delta_k D_k + \varepsilon$$

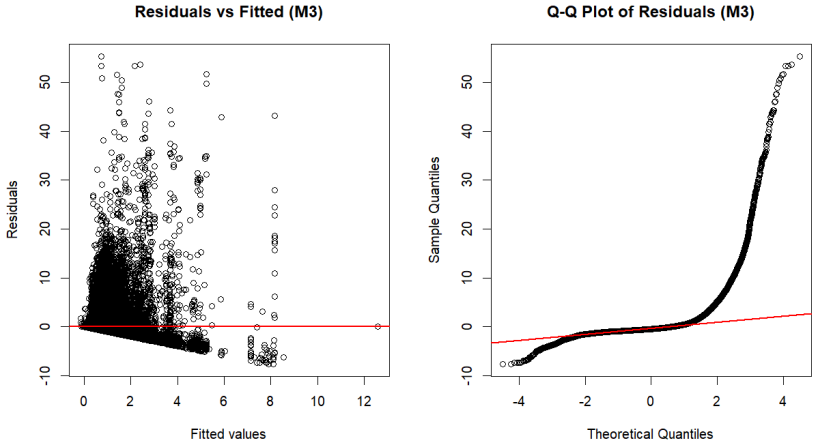

Figure 8: Residual diagnostics for Model 3.

Model 3 extends the previous specifications by incorporating geographic fixed effects (*State*) and categorical weather conditions in addition to numerical weather variables. The linear weather coefficients for temperature, humidity, pressure, and visibility remain statistically significant and negative based on individual t-tests, while wind speed is not statistically significant. This indicates that the primary weather effects are robust to the inclusion of categorical controls.

A substantial number of state indicators are statistically significant, suggesting meaningful geographic heterogeneity in accident distance across U.S. states. Similarly, several weather condition categories—particularly snow-, rain-, and fog-related conditions—exhibit strong and statistically significant effects, highlighting the importance of contextual environmental factors beyond continuous weather measures.

The overall F-test is highly significant ($F = 63.51$, $p < 0.001$), rejecting the null hypothesis that all slope coefficients are jointly equal to zero. The inclusion of categorical variables results in a marked improvement in explanatory power, increasing R^2 to approximately 0.051. Although residual non-normality persists, Model 3 represents a meaningful gain in model fit relative to purely numerical specifications.

3.4 Model 4 (M4): Weather + Infrastructure

$$\text{Distance} = \beta_0 + \sum_{j=1}^{p} \beta_j X_j + \sum_{m=1}^{r} \theta_m I_m + \varepsilon$$

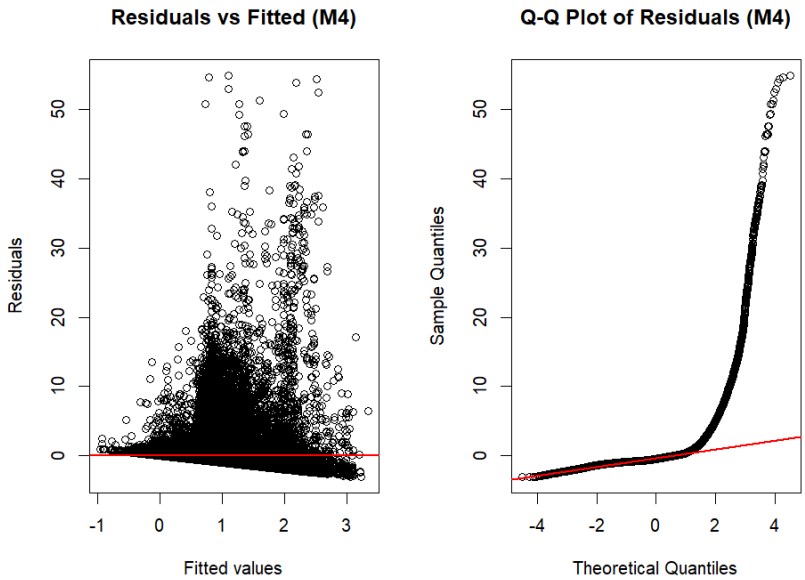

Figure 9: Residual diagnostics for Model 4.

Model 4 augments the baseline weather specification by incorporating binary road infrastructure indicators, including traffic signals, junctions, crossings, railways, stops, and stations. Among the numerical weather variables, humidity, atmospheric pressure, and visibility remain statistically significant with negative coefficients, while temperature is no longer statistically significant. Wind speed exhibits a small but statistically significant positive effect based on the individual t-test.

Several infrastructure variables are highly significant. The presence of traffic signals, crossings, stops, and stations is associated with shorter accident distances, whereas junctions are associated with longer affected road segments. The railway indicator is not statistically significant, suggesting no measurable effect after controlling for other factors.

The overall F-test strongly rejects the null hypothesis that all slope coefficients are jointly zero ($p < 0.001$), confirming the joint relevance of the included predictors. However, the explanatory power of the model remains limited, with R^2 of approximately 0.035. This indicates that while infrastructure characteristics contribute additional information, they explain only a modest share of the variation in accident distance.

3.5 Model 5 (M5): Full Linear Model

$$\text{Distance} = \beta_0 + \sum_{j=1}^{p} \beta_j X_j + \sum_{m=1}^{r} \theta_m I_m + \sum_{k=1}^{q} \delta_k D_k + \varepsilon$$

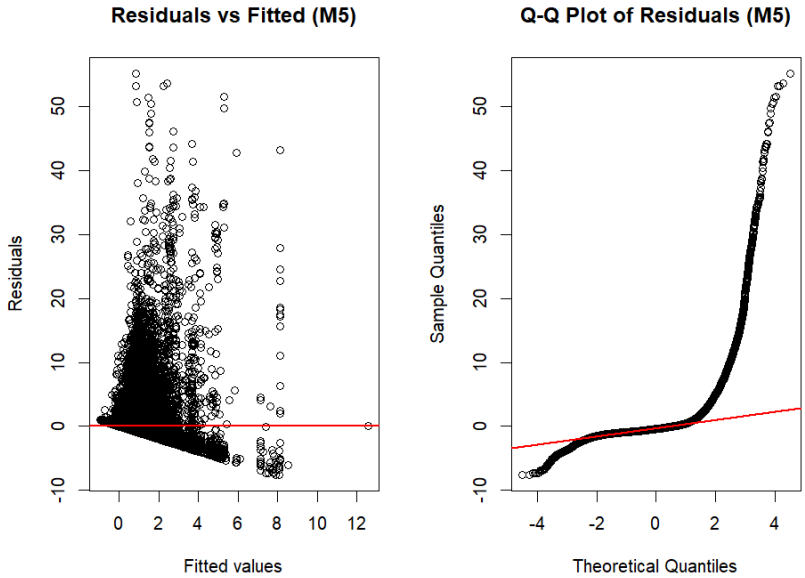

Figure 10: Residual diagnostics for Model 5.

Model 5 represents the most comprehensive linear specification on the original response scale, combining numerical weather variables, road infrastructure indicators, state fixed effects, and detailed weather condition categories.

Among the numerical weather variables, temperature, humidity, atmospheric pressure, and visibility remain statistically significant with negative coefficients, indicating that adverse or extreme weather conditions are associated with shorter affected road distances. Wind speed, however, is not statistically significant based on its t-test, suggesting no independent effect after controlling for other variables.

Several infrastructure variables are strongly significant. The presence of traffic signals, crossings, and stops is associated with shorter accident distances, while junctions are linked to longer affected segments. The railway indicator remains statistically insignificant, implying no measurable effect once other factors are accounted for.

Geographic effects play a substantial role in this model. Many state indicators are statistically significant, highlighting systematic regional differences in accident distance. In particular, some states exhibit markedly larger average accident distances relative to the reference state. Similarly, numerous weather condition categories—especially snow-related and heavy precipitation conditions—show strong and statistically significant effects, confirming the importance of extreme weather events.

The overall F-test strongly rejects the null hypothesis that all slope coefficients are jointly zero ($p < 0.001$), indicating that the model is statistically meaningful as a whole. Model 5 achieves the highest explanatory power among linear models on the original scale, with $R^2 \approx 0.062$. Although the proportion of explained variance remains modest, this improvement reflects the added contribution of geographic and contextual variables beyond weather alone.

3.6 Model 6 (M6): Log-Transformed Response

$$\log(\text{Distance} + 1) = \beta_0 + \sum_{j=1}^{p} \beta_j X_j + \sum_{m=1}^{r} \theta_m I_m + \sum_{k=1}^{q} \delta_k D_k + \varepsilon$$

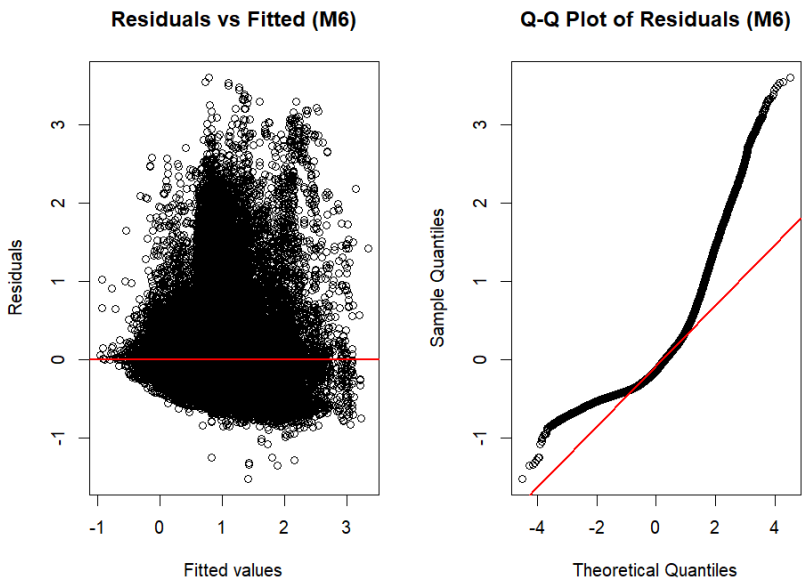

Figure 11: Residual diagnostics for Model 6.

Model 6 applies a logarithmic transformation to the response variable, $\log(\text{Distance} + 1)$, to address strong right-skewness and heteroscedasticity observed in previous linear models.

Most numerical weather variables are statistically significant according to their t-tests. Temperature, humidity, atmospheric pressure, and visibility exhibit negative coefficients, indicating that adverse weather conditions are associated with proportionally shorter affected road distances. In contrast, wind speed has a positive and statistically significant effect, suggesting that stronger winds are linked to relatively longer accident impacts on a multiplicative scale.

Infrastructure indicators remain highly significant. The presence of traffic signals and crossings is associated with a substantial reduction in the log-distance of accidents, while junctions are linked to longer affected road segments. These effects are more precisely estimated compared to models on the original scale, reflecting improved model stability after transformation.

Geographic variation is pronounced in this specification. A large majority of state indicators are statistically significant, highlighting persistent regional differences in accident distance even after controlling for weather and infrastructure. The magnitude of these coefficients is naturally interpreted in percentage terms due to the logarithmic response.

The overall F-test strongly rejects the null hypothesis that all slope coefficients are jointly zero ($p < 0.001$), confirming the statistical significance of the model. Model 6 achieves a substantially higher explanatory power compared to previous specifications, with $R^2 \approx 0.082$, and exhibits markedly improved residual behavior, making it the most reliable linear model in the analysis.

3.7 Model 7 (M7): Log Response with State and Weather Conditions

$$\log(\text{Distance} + 1) = \beta_0 + \sum_{j=1}^{p} \beta_j X_j + \sum_{k=1}^{q} \delta_k D_k + \varepsilon$$

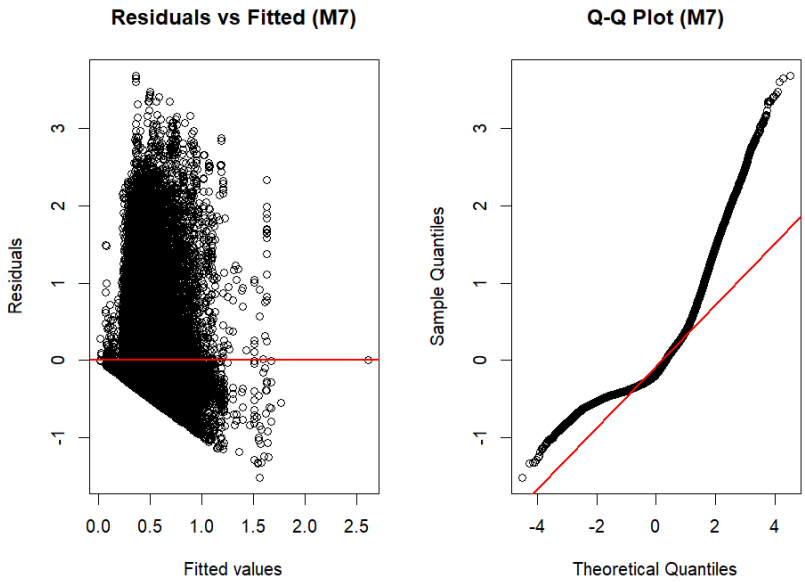

Figure 12: Residual diagnostics for Model 7.

Model 7 extends the log-transformed specification by incorporating both geographic (State) and categorical weather condition variables alongside numerical weather predictors. The response variable remains $\log(\text{Distance} + 1)$, allowing effects to be interpreted on a multiplicative scale.

Among numerical weather variables, humidity, atmospheric pressure, and visibility remain highly statistically significant according to their t-tests, with negative coefficients indicating shorter accident impact distances under adverse conditions. Temperature and wind speed are no longer statistically significant once regional and weather condition controls are introduced, suggesting their effects are absorbed by these categorical factors.

Geographic heterogeneity continues to play a major role. A large proportion of state indicators are statistically significant, with positive coefficients for many states, highlighting persistent regional differences in accident distance even after accounting for weather conditions. These state-level effects remain one of the dominant contributors to explanatory power in the model.

Several weather condition categories are statistically significant. Severe conditions such as heavy snow, sleet, wintry mix, and thunder-related events are associated with substantially larger accident distances on the log scale, while lighter precipitation and cloudy conditions tend to reduce accident impact length relative to the baseline category.

The overall F-test strongly rejects the null hypothesis that all slope coefficients are jointly zero ($p < 0.001$), confirming the joint significance of the model. However, the explanatory power of Model 7 ($R^2 \approx 0.056$) is lower than that of Model 6, indicating that the inclusion of a large number of categorical weather conditions increases model complexity without improving

predictive performance.

3.8 Model 8 (M8): Log Response with Polynomial Weather Effects

$$\log(\text{Distance} + 1) = \beta_0 + \sum_{j=1}^{p} \beta_j X_j + \sum_{j=1}^{p} \gamma_j X_j^2 + \sum_{k=1}^{q} \delta_k D_k + \varepsilon$$

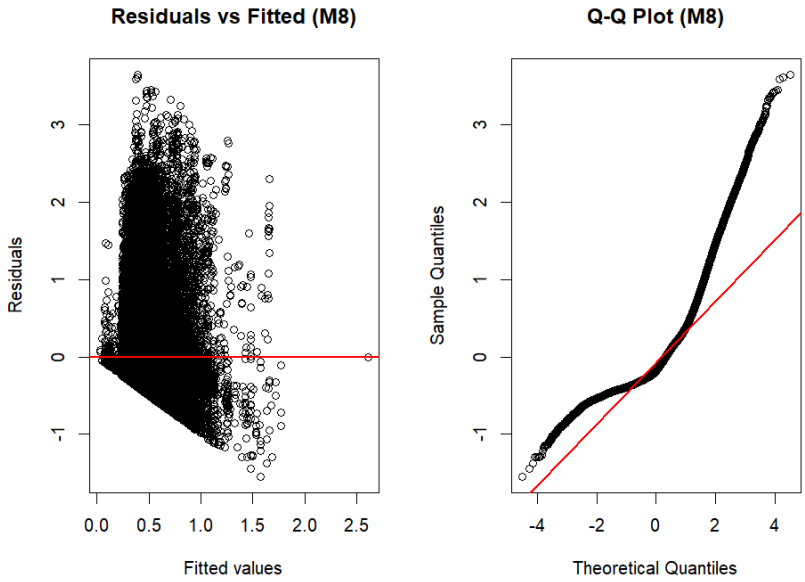

Figure 13: Residual diagnostics for Model 8.

Model 8 extends the log-transformed specification by introducing quadratic terms for numerical weather variables, allowing for nonlinear effects while retaining state and weather condition controls. The response variable remains $\log(\text{Distance} + 1)$.

Both the linear and squared terms of temperature, humidity, and atmospheric pressure are highly statistically significant according to their t-tests. The opposing signs of linear and quadratic coefficients indicate nonlinear relationships, suggesting diminishing or accelerating effects of weather severity on accident distance. Visibility also exhibits a significant nonlinear pattern, while wind speed and its squared term are not statistically significant once other predictors are controlled for.

Geographic effects remain strong and consistent with earlier models. A large majority of state indicators are statistically significant, confirming persistent regional heterogeneity in accident impact length even after accounting for nonlinear weather effects.

Several severe weather conditions, including heavy snow, sleet, wintry mix, and thunder-related events, are associated with substantially larger accident distances on the log scale. Lighter precipitation and cloudy conditions tend to reduce accident distance relative to the baseline category.

The overall F-test strongly rejects the null hypothesis that all slope coefficients are jointly zero ($p < 0.001$), confirming the joint significance of the model. However, despite its increased complexity, Model 8 achieves only a modest improvement over Model 7, with $R^2 \approx 0.058$. This

suggests that the inclusion of polynomial weather terms adds limited explanatory power and may introduce unnecessary complexity without substantial gains in predictive performance.

3.9 Model 9 (M9): Mixed Transformations Model

$$\text{Distance} = \beta_0 + \sum_{j=1}^{p} \beta_j X_j + \gamma_1 Z_1^2 + \gamma_2 \log(Z_2) + \gamma_3 \log(Z_3) + \sum_{k=1}^{q} \delta_k D_k + \varepsilon$$

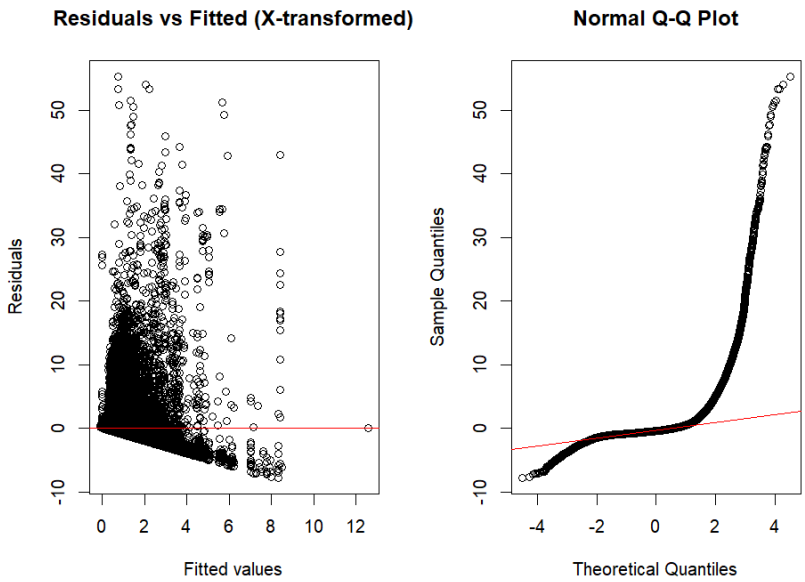

Figure 14: Residual diagnostics for Model 9.

Model 9 explores a mixed transformation strategy by combining linear, quadratic, and logarithmic transformations of numerical predictors while retaining categorical controls for state and weather conditions. Unlike the previous models, the response variable is specified on its original scale, Distance, while selected predictors are transformed to capture nonlinear effects.

Weather-related variables remain largely significant according to their t-tests. Temperature and humidity exhibit negative linear effects, indicating shorter accident distances under harsher conditions. Atmospheric pressure shows a strong nonlinear relationship, with both the linear and squared terms being highly significant, suggesting curvature in the pressure–distance association. Log-transformed visibility has a significant negative effect, indicating that reduced visibility is associated with longer affected road segments, whereas log-transformed wind speed is not statistically significant.

Geographic heterogeneity persists in this specification. Many state indicators are statistically significant, confirming substantial regional variation in accident distance even after accounting for nonlinear and transformed weather effects. The magnitude and direction of these coefficients are broadly consistent with earlier linear models.

Several severe weather categories, including heavy snow, sleet, wintry mix, and thunder-related conditions, are associated with markedly larger accident distances. In contrast, fair and cloudy conditions tend to reduce accident distance relative to the baseline category.

The overall F-test strongly rejects the null hypothesis that all slope coefficients are jointly zero ($p < 0.001$), confirming the joint statistical significance of the model. However, Model 9 does not improve explanatory power relative to the log-transformed specifications, achieving only $R^2 \approx 0.053$. Residual diagnostics indicate limited improvement in distributional behavior compared to earlier models, suggesting that combining multiple transformation types on the original response scale adds complexity without yielding substantial gains. Consequently, this model is less effective than the log-transformed alternatives.

4 Model Comparison

Table 1: Comparison of Model Performance Based on R^2 and Mean Squared Error

Model	Response	R^2	Adj. R^2	MSE
M1	Distance	0.023	0.023	4.3208
M2	Distance	0.025	0.025	4.3126
M3	Distance	0.051	0.050	4.1956
M4	Distance	0.035	0.035	4.2684
M5	Distance	0.062	0.061	**4.1503**
M6	$\log(D+1)$	0.082	0.082	4.3795
M7	$\log(D+1)$	0.056	0.055	4.3780
M8	$\log(D+1)$	0.058	0.058	4.3692
M9	Distance	0.053	0.052	4.1875

Table 1 compares all nine models using explanatory power (R^2) and predictive accuracy (MSE).

Models M1 and M2 show limited performance, indicating that numerical weather variables alone—linear or quadratic—explain little variation in accident distance. The inclusion of categorical variables in M3 leads to a clear improvement, while infrastructure variables in M4 provide only modest gains.

Model M5 achieves the lowest MSE among all models on the original distance scale, making it the best-performing model for prediction. Log-transformed models (M6–M8) improve goodness-of-fit and residual behavior but yield higher MSE values when evaluated on the original scale.

The mixed-transformation model (M_xt) improves upon simple linear specifications but does not outperform M5. Overall, M5 is preferred for prediction, while M6 is better suited for inference and diagnostics.

4.1 Confidence Intervals for the Final Model

To quantify the uncertainty associated with the estimated coefficients in the final linear specification (Model 5), 95% confidence intervals were computed for all parameters. For each coefficient β_j, the confidence interval is given by

$$\hat{\beta}_j \pm 1.96 \cdot \mathrm{SE}(\hat{\beta}_j),$$

where $\mathrm{SE}(\hat{\beta}_j)$ denotes the standard error of the estimate.

Table 2 reports selected confidence intervals for the main numerical and infrastructure-related predictors in Model 5.

Table 2: 95% Confidence Intervals for Selected Coefficients (Model 5)

Variable	2.5%	97.5%
Intercept	5.630	6.518
Temperature (°F)	-0.00343	-0.00115
Humidity (%)	-0.00436	-0.00302
Pressure (in.)	-0.17338	-0.14669
Visibility (mi.)	-0.02670	-0.01379
Wind Speed (mph)	-0.00179	0.00276
Traffic Signal	-0.54617	-0.46039
Junction	0.09017	0.17015
Crossing	-0.37167	-0.27506
Railway	-0.16133	0.09768
Stop	-0.61271	-0.47124

The confidence intervals provide additional insight beyond statistical significance. Coefficients whose intervals do not include zero indicate robust effects at the 5% significance level. For example, temperature, humidity, pressure, and visibility all exhibit strictly negative intervals, confirming a consistent negative association with accident distance.

In contrast, the confidence interval for wind speed includes zero, indicating that its effect is not statistically distinguishable from zero once other covariates are controlled for. Among infrastructure variables, traffic signals and crossings show clearly negative intervals, while junctions are associated with longer affected distances.

The width of several intervals—particularly for state and weather-condition indicators (not shown)—reflects substantial heterogeneity and uncertainty, consistent with the limited explanatory power of the model.

5 Conclusion: Interpretation and Data Limitations

The sequence of nine regression models was constructed based on increasing theoretical complexity and practical relevance. Initial models focused on numerical weather variables, as weather conditions are commonly associated with traffic accidents and may influence accident severity and response. Subsequent models incorporated nonlinear effects, infrastructure indicators, geographic variation, and categorical weather conditions to capture heterogeneity across locations and environments. Logarithmic and mixed transformations were applied to address skewness, heteroscedasticity, and potential nonlinear relationships in the response variable.

Despite these theoretically motivated specifications, all models exhibit relatively low explanatory power. This outcome is largely driven by the nature of the response variable, *Distance*, which represents the physical length of the roadway affected by an accident. Accident

distance is strongly influenced by structural and operational factors such as road type, number of lanes, traffic volume, and traffic management decisions made by authorities.

In open or rural areas, road closures may extend over long distances even when traffic disruption is limited. In contrast, in dense urban environments or during peak traffic hours, an accident may affect only a short road segment while causing substantial congestion. These dynamics are not directly captured by the available predictors.

The dataset primarily includes environmental, infrastructural, and contextual variables, but lacks key operational information such as real-time traffic conditions, enforcement strategies, and incident response protocols. As a result, although many coefficients are statistically significant due to the large sample size, the models are unable to explain a substantial share of the variation in accident distance. This limitation explains the consistently low R^2 values observed across all specifications.